

Pricing and Market Making in Short-Horizon Crypto Prediction Markets

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Abstract

Short-horizon binary contracts connect a continuous underlying price to a discrete terminal payoff. This study examines pricing structure, order-book measurement and market-making policy evaluation in historical Polymarket BTC five-minute up/down markets. A Brownian-probit model describes broad within-slot variation in market mids, but structural fit does not identify an individual participant's strategy or establish profitable forecasting. Execution modeling introduces additional constraints: venue-specific feed levels can manufacture signals when merged incorrectly, and hypothetical fills cannot occur before order activation. A latency-corrected static front-quoting experiment produced fresh-data expected value of -0.9845 cents per eligible decision moment, with a 90% slot-cluster interval of $[-1.626, -0.364]$, across 312 moments in 193 slots. The estimate is negative even with an unobserved additional fill cost set to zero. Positive results for other replay policies remain model-dependent, while defective paper runs support operational diagnosis rather than economic conclusions. The accompanying repository provides reference components and verification of a decision rule from aggregate evidence; it does not independently reconstruct the original raw-data estimates.

1. Research question and contribution

The motivating question is how a compact description of market prices translates into an executable policy. A model can reproduce the broad shape of binary prices while omitting the conditional information contained in fills, queue priority and the response of other traders to an order. Those omissions matter most when expected spread income is small relative to adverse selection.

The research programme covers three connected questions. First, how well does a simple binary pricing map describe observed market mids? Second, which parts of the observed feed/book relationship survive measurement controls? Third, do policies built from those relationships retain favorable economics after explicit activation, fill and inventory assumptions?

The contribution is an empirical and engineering case study, not a new general pricing theorem. The selected experiments were conducted during June–July 2026 under the study's point-settlement specification. Earlier positive results and later corrections are retained as a chronological record. The

central policy conclusion is a rejection within a defined model and historical sample, not a claim that profitable market making is impossible.

2. Instrument and observation model

The studied contracts represent complementary Up and Down claims on a five-minute BTC price window. The research treats their outcome as a comparison of reference observations at the start and end of the window. A complementary pair has terminal payoff one dollar. The study's continuous-price model assigns zero probability to an exact boundary tie; the actual tie convention belongs to the individual contract specification. The public package does not yet include a complete archived rule snapshot.

The contractual reference, external spot feeds and contract book are distinct observations. A feed can lead a displayed reference without providing executable arbitrage. A level difference between venues can also be stable without being a forecast. The price model's feed-self anchor describes a market-mid relationship and must not be silently substituted for the contractual resolution anchor.

Records used by the broader programme include exchange ticks, book updates and trades. Collection and observation timestamps require separate treatment. Offline studies use as-of values and an explicit clean-data index. The original data audit identified CLOB WebSocket freezes up to 32 seconds while external feeds continued to move. Such intervals can resemble stable quoting opportunities unless excluded. The public reference implementation illustrates masking; the full raw recordings are not distributed with this manuscript [R1].

3. Pricing structure

Consider an idealized driftless diffusion with constant volatility scale sigma. For displacement x from the anchor and remaining time τ , the binary probability map is

$$p(x, \tau) = \Phi\left(\frac{x}{\sigma\sqrt{\tau}}\right).$$

The transformation

$$z = \Phi^{-1}(p)\sqrt{\tau}$$

is linear in x under this specification. The historical pricing report records median within-slot R-squared of approximately 0.92 for transformed regressions. This is an in-sample statistic in transformed space, not a 92% directional prediction rate. A specification frozen using earlier logs subsequently reproduced held-out market mids at approximately six ticks RMSE; one tick in this study is one cent. These two summaries describe different evaluation views of the same pricing model [R2].

The public package supplies the transform and inversion, but not the original regression logs. Accordingly these figures are author-reported historical estimates, not independently reproduced results of the public demonstration. They also do not isolate a dominant maker: market mids aggregate multiple participants and can be consistent with several underlying strategies.

The historical comparison found better realized-outcome calibration in market prices when the model and market disagreed. This restricted the model's use to structural interpretation and timing research. It did not support treating model-market disagreement as a profitable signal. The result does not establish universal market efficiency outside the measured samples.

Under the ideal constant-parameter model, the induced price diffusion is

$$\sigma_p(p, \tau) = \frac{\phi(\Phi^{-1}(p))}{\sqrt{\tau}}.$$

Its interior-price behavior near expiry explains why a fixed underlying move can cause large changes in a binary probability. It does not mean every realized path has diverging risk: probabilities may move toward a boundary. Re-estimating volatility or using a forecast proxy also changes the assumptions needed for a martingale interpretation. A dynamic volatility heuristic alone does not guarantee it.

4. Measurement and execution

4.1 Feed construction

A merged last-value stream alternates between whichever venue last reported. With a persistent inter-venue price offset, it creates a sawtooth pattern even when each venue's own path is smooth. Matched controls exposed this artifact in the original event studies. Subsequent signal construction uses Binance alone. This differs from constructing a synchronized composite covariate for a pricing regression: the sampling operation, not merely the number of sources, determines the artifact [R1].

4.2 Queue access

Level-2 observations do not reveal the queue rank of a counterfactual order. The research therefore walks hypothetical orders through visible depth and subsequent prints. Trades-only depletion and depletion that also credits unexplained size reductions are sensitivity scenarios. They are not guaranteed bounds on real fills: hidden liquidity, replenishment, priority and changes caused by the additional order are unobserved.

The historical maker decision cell uses 50 ms activation and 50 ms cancellation. Maker POST measurements were 24–50 ms, while cancel median measurements were 23–50 ms with a longer tail under load. Older 250–330 ms figures concern the taker path. A separate paper campaign used

incorrect execution assumptions; that comparison was subsequently invalidated [R3]. These are historical engineering measurements, not advertised current venue service levels.

An order decided at t cannot consume prints before t plus activation delay. In the initial front experiment, this distinction was omitted. Correcting it removed a substantial part of the apparent edge. A cancel request likewise leaves a period of exposure before modeled removal.

5. Inventory and expected value

Let U denote the Up payoff. Holding q_U Up shares and q_D Down shares gives terminal value

$$q_D + (q_U - q_D)U.$$

With net inventory $q = q_U - q_D$ and conditional outcome probability p , the conditional variance is

$$q^2 p(1 - p).$$

Time does not appear explicitly, but p changes with time and information. This expression must not be interpreted as constant unconditional risk until settlement. If both shares from a split pair are sold at a and b , the paired trading result before costs is $a + b - 1$. If only one is sold, the remaining inventory retains outcome risk. Historical rebates enter separately from trading income.

The research represents policy choices using states, actions, transition components and rewards. The implementation is an MDP-inspired generative expected-value decomposition rather than a solved Bellman control problem. It distinguishes a policy's observable fill-conditioned drift from an additional unobserved cost parameter. With non-negative exposure weight w ,

$$EV(\alpha) = EV(0) - w\alpha, \quad \alpha \geq 0.$$

A negative $EV(0)$ rejects the candidate for all non-negative values of this additional cost within the model. This is a sensitivity argument conditional on the other measurement assumptions. It does not independently prove that the queue proxy, eligibility rules or estimated drift are correct.

Inventory-aware quoting has a substantial existing literature. Avellaneda and Stoikov [L1] provide a continuous-price market-making framework; its assumptions should be re-derived for the payoff and execution problem being studied here. Adverse selection from informed trading provides a separate reason for spreads and fill-conditioned losses, as modeled by Glosten and Milgrom [L2]. Neither work implies that a particular empirical implementation here is profitable.

6. Static front-quoting experiment

The D12 front-calm cell uses clean full slots, decision times 10–200 seconds after the window opens, and paired samples every 50 book events. Calm is defined from Binance's as-of 1.5-second displacement with absolute magnitude below three dollars. Two-sided improvement requires a spread of at least three cents, so improving each side by one tick does not create a locked quote. These are historical definitions of the evaluated candidate, not parameters selected for current trading [R3].

Fills start after 50 ms activation. The front proxy uses subsequent side-matched prints while the improved level remains available, capped by a primary horizon of ten seconds. The recorded book's arrival at the improved level closes this opportunity window. Because the hypothetical improved level did not necessarily exist on tape, its fill process has counterfactual uncertainty. Dropouts and missing or stale future marks remove affected samples. The public verifier does not rerun that event-level construction.

The original protocol includes chronological evaluation, queue and regime permutation controls, slot-cluster bootstrap intervals and stability checks. Protocol existence alone is not evidence that every individual study passed every check. The public D12 summary includes the selected control outputs, but confidence intervals and placebo estimates are reported from the upstream estimator.

SAMPLE OR SENSITIVITY	EV(o), CENTS PER ELIGIBLE MOMENT
Discovery block A	+0.3383
Discovery block B	+0.4325
Grown fresh block F	−0.9845
Fresh 90% slot-cluster interval	[−1.626, −0.364]
Chronological final 30%	−0.9989
Fresh result after the most favorable single-slot omission	−0.8792

The fresh result covers 312 eligible moments in 193 slots. Moments are not treated as independent observations for interval construction; the reported uncertainty unit is the slot. The chronological split and fresh block are related checks and should not be counted as independent replications. The grown fresh result supersedes earlier estimates on a smaller fresh sample [R4].

The interval is entirely negative at alpha zero. A single omitted slot does not restore the sign. The resulting decision is to reject this static policy within its represented assumptions. The public command `python -B estimators/d12_public_verifier.py --check` verifies that this decision agrees with the committed aggregate input. It does not rebuild the sampling distribution.

7. Other policies and the limits of paper comparison

Inventory rebalancing and decision-time EV policies produced positive replay comparisons under their declared controls. Those findings remain model-dependent upper bounds because visible historical flow does not reveal the response to our own orders. They are part of the research record, but no live-profitability estimate is inferred from them [R3].

A separate diagnostic aligned replay with a six-slot stress window observed by a paper engine. Replay credited approximately +\$21 while the paper engine reported approximately −\$31. Investigation identified cold-start regime blindness, stale-feed handling and inventory/loss-limit defects. The comparison demonstrates an implementation and model-coverage gap. Because the paper run was defective, it supports no economic conclusion about a correctly functioning strategy [R5].

An earlier 157-slot A/B comparison is a distinct experiment. Subsequent inspection found incorrect execution assumptions and invalidated its economic interpretation. The corrected research record therefore does not retain its old inconclusive label as evidence from a clean comparison. Aligning time windows is necessary for comparison but insufficient when engine mechanics differ.

8. Reproducibility and limitations

The repository offers three inspectable layers: mathematical reference functions, a synthetic event walkthrough, and a deterministic decision-rule verifier over aggregate output. The component suite contains 62 tests. These checks establish implementation behavior and agreement with stored evidence; they do not authenticate the original observations or independently validate empirical estimates.

The main limitations are unpublished raw-data reconstruction, inferred queue priority, counterfactual execution response, limited historical regimes and incomplete archived instrument rules. Multiple experiments informed the research sequence, so a clean fresh split should not be mistaken for a fully preregistered programme with no researcher adaptation. The final policy is specified explicitly to avoid extending its rejection to untested families.

The owner reports a subsequent change toward averaging in settlement. This paper makes no exact transition-date or weighting claim and does not transfer point-settlement results to that payoff. Archived primary rule evidence would strengthen the historical specification without requiring a new strategy study. Further empirical reconstruction could also improve independent reproducibility.

9. Conclusion

The historical price process admitted a useful compact description, while executable policy value depended on additional assumptions about activation, queue access, selection and inventory. Corrected measurement rejected the specified static front policy on fresh data. Other positive replay findings and broken paper comparisons did not establish live profitability. The resulting research package separates structural modeling, empirical summaries and operational diagnosis so that each can be reviewed at the level of evidence it actually provides.

References and research artifacts

- [L1] Avellaneda, M. and Stoikov, S. (2008). *High-frequency trading in a limit order book*. Quantitative Finance, 8(3), 217–224. [Author-hosted paper](#).
- [L2] Glosten, L. R. and Milgrom, P. R. (1985). *Bid, Ask, and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders*. Journal of Financial Economics, 14(1), 71–100. [Columbia research record](#).
- [R1] Author research records, Do/D4 data and feed studies. Public method summary: [queue and fill mechanics](#). Raw recordings not included.
- [R2] Author pricing research report, June 2026. Public summary: [market pricing model](#). Original regression logs not included.
- [R3] Author D12 methods and chronological research log, July 2026. Public summaries: [EV decomposition](#) and [experiment revisions](#).
- [R4] [D12 aggregate estimator output](#) and [verified decision summary](#). The summary records its input SHA-256.
- [R5] Author D16 and A/B audits, July 2026. Public summary: [replay and paper coverage](#). Original paper journals not included.

Author and development note

The research is by Eren Ege Çelik. AI coding and writing assistants were used in implementation, documentation and preparation of this working manuscript. The manuscript is offered for feedback and has not been peer reviewed. Questions about the model assumptions, empirical identification and reproduction boundary are especially welcome.